



US012410700B2

(12) **United States Patent**
Maus et al.

(10) **Patent No.:** **US 12,410,700 B2**

(45) **Date of Patent:** **Sep. 9, 2025**

(54) **METHOD OF DRILLING A WELLBORE TO A TARGET**

(71) Applicant: **Magnetic Variation Services, LLC**,
Denver, CO (US)

(72) Inventors: **Stefan Maus**, Boulder, CO (US); **Marc Ernest Willerth**, San Luis Obispo, CA (US); **Jarrod Shawn Deverse**, Greenwood Village, CO (US)

(73) Assignee: **Magnetic Variation Services, LLC**,
Denver, CO (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 545 days.

(21) Appl. No.: **17/808,527**

(22) Filed: **Jun. 23, 2022**

(65) **Prior Publication Data**

US 2022/0325616 A1 Oct. 13, 2022

Related U.S. Application Data

(63) Continuation of application No. 16/097,788, filed as application No. PCT/US2017/032295 on May 11, 2017, now Pat. No. 11,473,420.

(60) Provisional application No. 62/408,622, filed on Oct. 14, 2016, provisional application No. 62/335,078, filed on May 12, 2016.

(51) **Int. Cl.**
E21B 47/022 (2012.01)
E21B 7/04 (2006.01)
E21B 44/00 (2006.01)
E21B 47/024 (2006.01)
E21B 47/04 (2012.01)
G01V 3/38 (2006.01)

(52) **U.S. Cl.**

CPC **E21B 47/022** (2013.01); **E21B 7/04** (2013.01); **E21B 44/00** (2013.01); **E21B 47/024** (2013.01); **E21B 47/04** (2013.01); **G01V 3/38** (2013.01); **E21B 2200/20** (2020.05)

(58) **Field of Classification Search**

USPC 702/9
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,405,808	B1	6/2002	Edwards et al.
7,957,946	B2	6/2011	Pirovolou
8,561,720	B2	10/2013	Edbury et al.
11,421,525	B2	8/2022	Maus et al.
11,473,420	B2	10/2022	Maus et al.
2002/0060570	A1	5/2002	Shirasaka et al.

(Continued)

OTHER PUBLICATIONS

U.S. Appl. No. 16/097,788, "Advisory Action", Jan. 25, 2022, 5 pages.

(Continued)

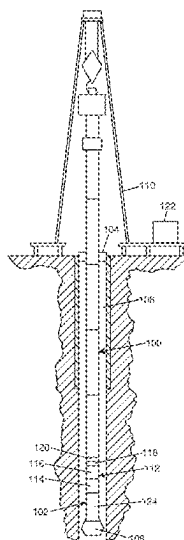
Primary Examiner — Paul D Lee

(74) *Attorney, Agent, or Firm* — Kilpatrick Townsend & Stockton LLP

(57) **ABSTRACT**

A method of drilling a wellbore to a target includes measuring attitudes at two adjacent survey stations along a wellbore using a downhole surveying tool. An actual change in wellbore position over a survey leg linking the two survey stations is determined. Corrections are applied to the measured attitude at one or both of the survey stations such that a modeled well path joining the survey stations reflects the determined actual change in wellbore position between the two survey stations.

24 Claims, 8 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

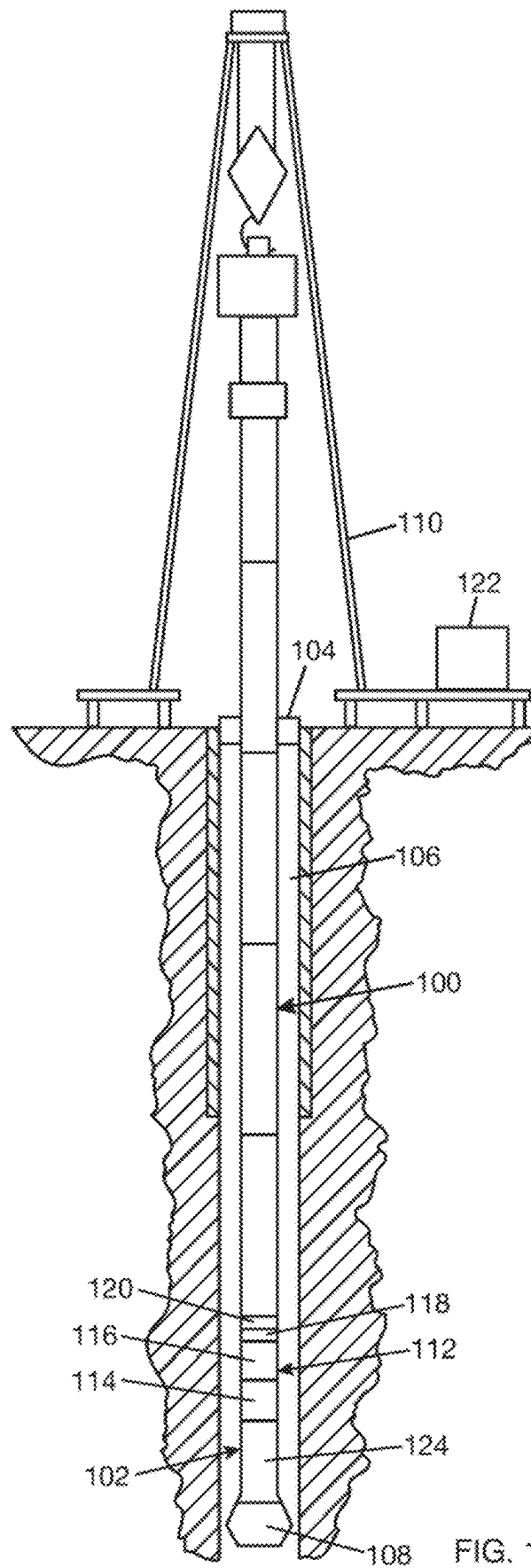
2009/0120690	A1 *	5/2009	Phillips	E21B 47/022	
					175/45
2012/0330551	A1	12/2012	Mitchell et al.		
2015/0240622	A1	8/2015	Bang et al.		
2015/0247397	A1 *	9/2015	Samuel	E21B 7/04	
					700/275
2015/0377004	A1 *	12/2015	Hornblower	E21B 7/06	
					175/24
2016/0032709	A1	2/2016	Ledroz et al.		
2019/0169979	A1 *	6/2019	Nguyen	G01C 1/00	

OTHER PUBLICATIONS

U.S. Appl. No. 16/097,788, "Final Office Action", Oct. 27, 2021, 23 pages.
 U.S. Appl. No. 16/097,788, "Non-Final Office Action", Jun. 8, 2021, 20 pages.
 U.S. Appl. No. 16/097,788, "Notice of Allowance", Mar. 24, 2022, 14 pages.
 U.S. Appl. No. 16/192,135, "Corrected Notice of Allowability", May 4, 2022, 2 pages.

U.S. Appl. No. 16/192,135, "Non-Final Office Action", Oct. 28, 2021, 16 pages.
 U.S. Appl. No. 16/192,135, "Notice of Allowance", Apr. 11, 2022, 10 pages.
 Australian Patent Application No. 2017263539, "First Examination Report", Feb. 22, 2022, 3 pages.
 Colombian Patent Application No. NC2018/0012148, "Office Action", Dec. 14, 2020, 20 pages.
 European Patent Application No. 17796894.8, "Extended European Search Report", Jan. 27, 2020, 8 pages.
 PCT/US2017/032295, "International Preliminary Report on Patentability", Nov. 22, 2018, 9 pages.
 PCT/US2017/032295, "International Search Report and Written Opinion", Aug. 16, 2017, 9 pages.
 AU2017263539, "Notice of Acceptance", Nov. 21, 2022, 3 pages.
 AU2017263539, "Second Examination Report", Sep. 5, 2022, 2 pages.
 CA3,022,843, "Office Action", Jun. 12, 2024, 6 pages.
 U.S. Appl. No. 16/097,788, "Notice of Allowance", Jul. 27, 2022, 8 pages.
 Mexican Patent Application No. MX/A/2018/013666, "Office Action", Aug. 30, 2022, 4 pages.
 CA3022843, "Office Action", Aug. 9, 2023, 6 pages.
 EP17796894.8, "Intention to Grant", Oct. 4, 2024, 7 pages.

* cited by examiner



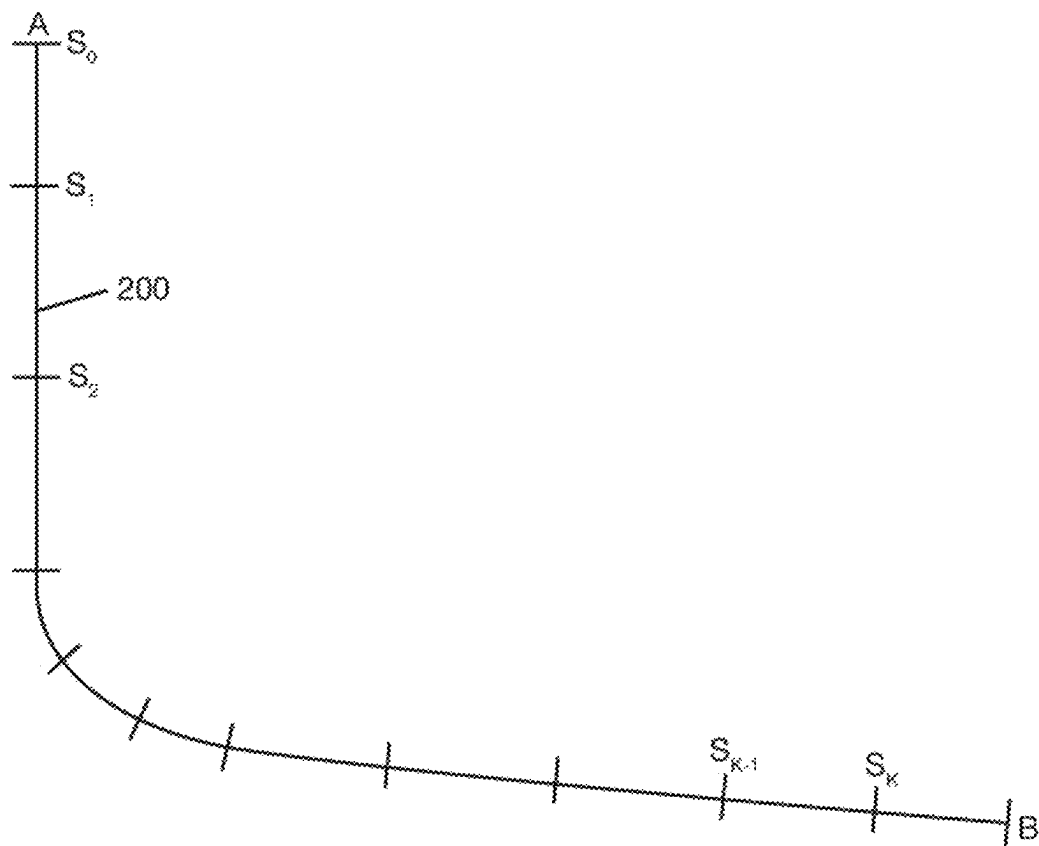


FIG. 2

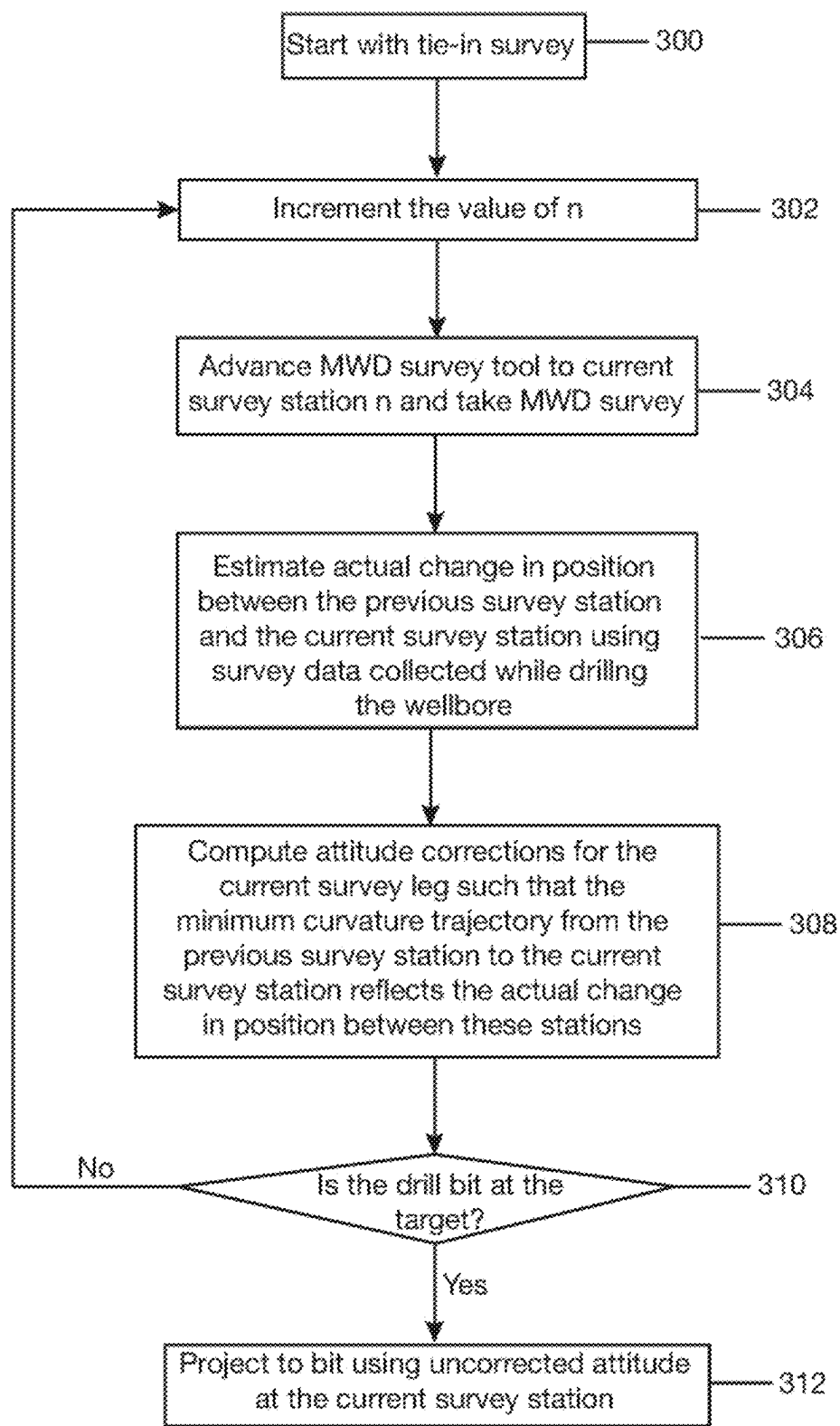


FIG. 3A

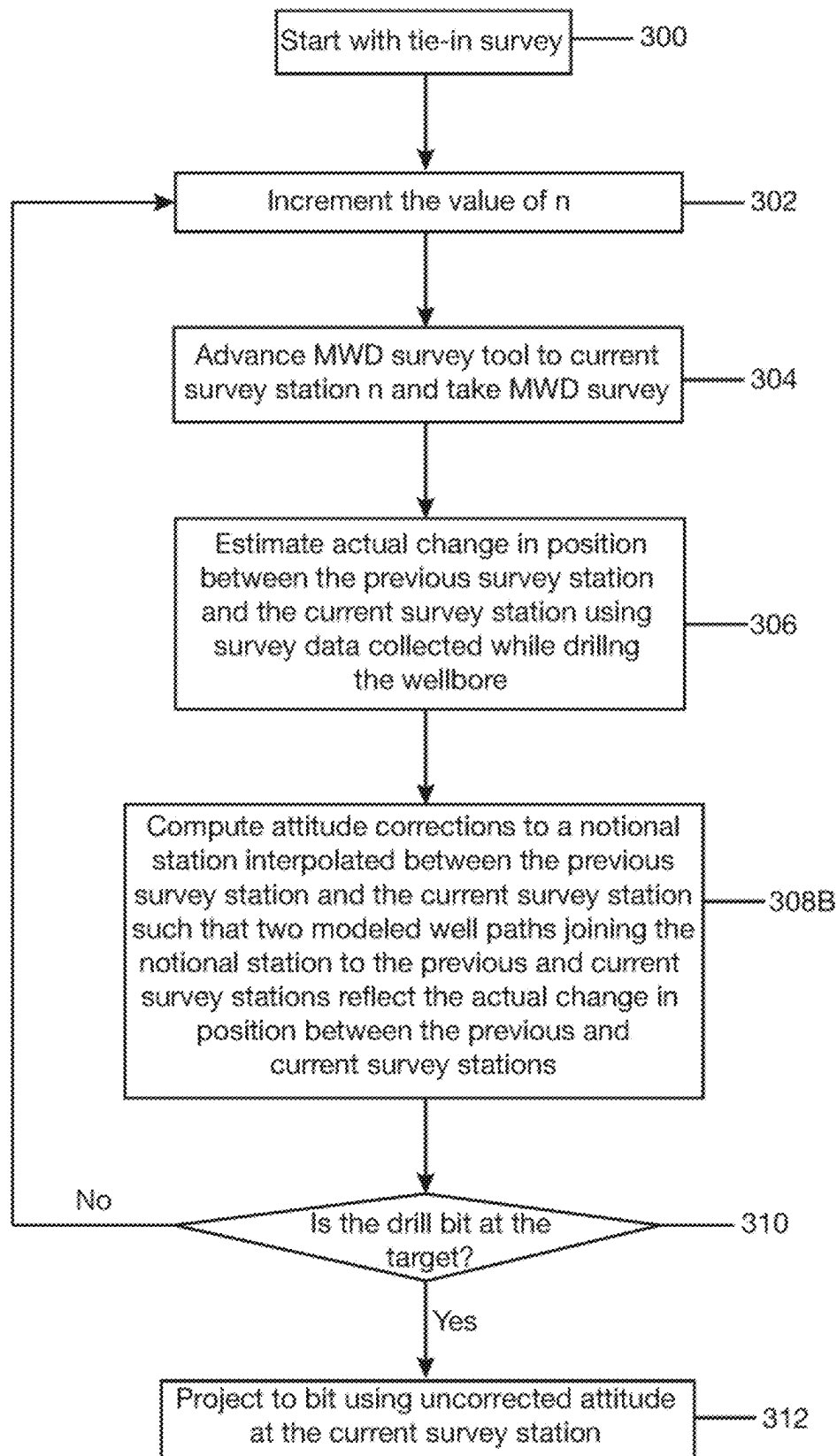


FIG. 3B

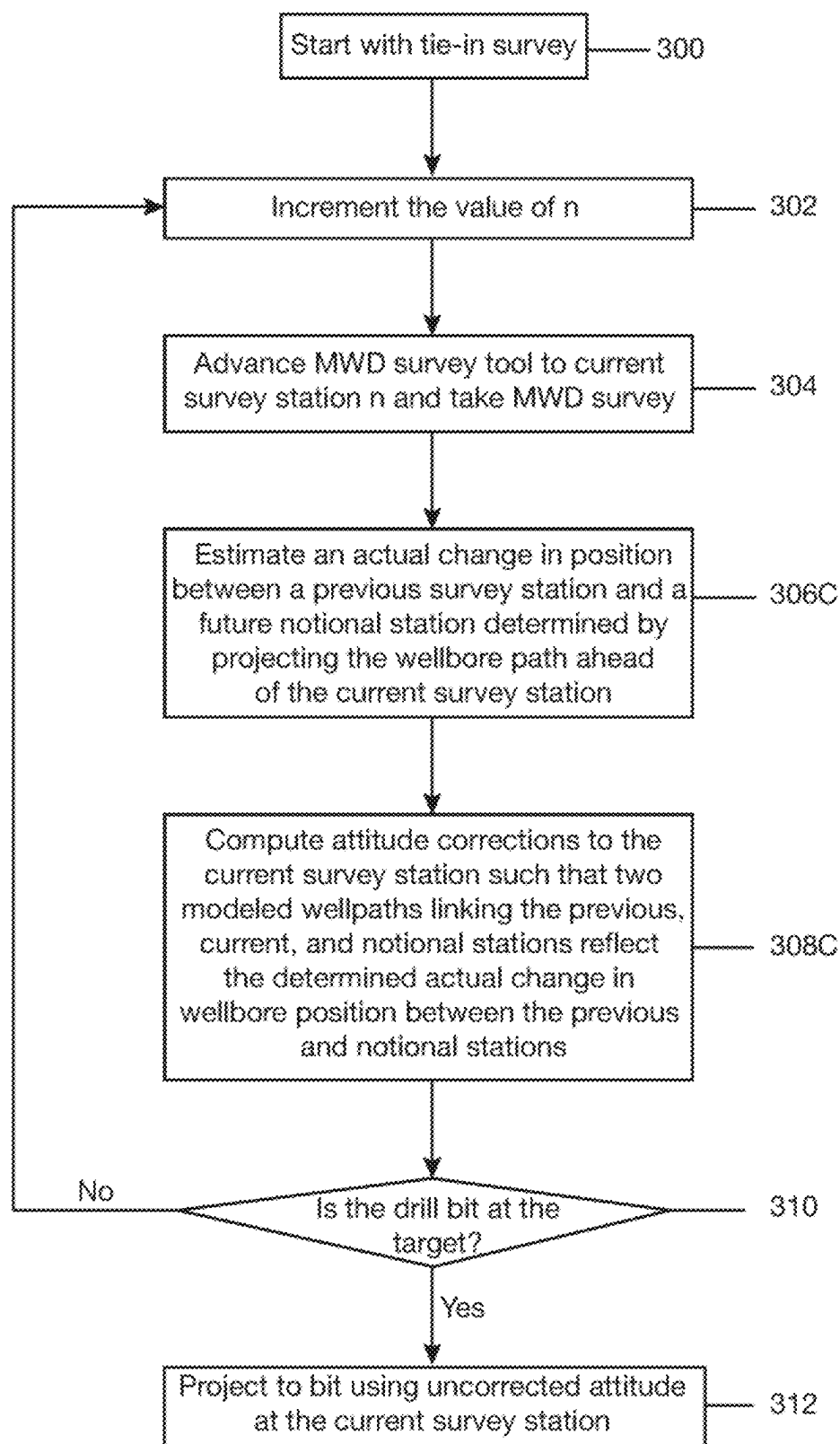
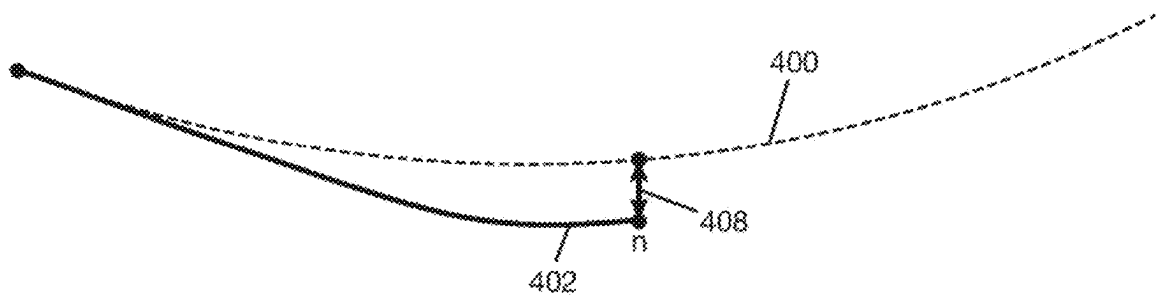


FIG. 3C



(PRIOR ART)
FIG. 4A



FIG. 4B

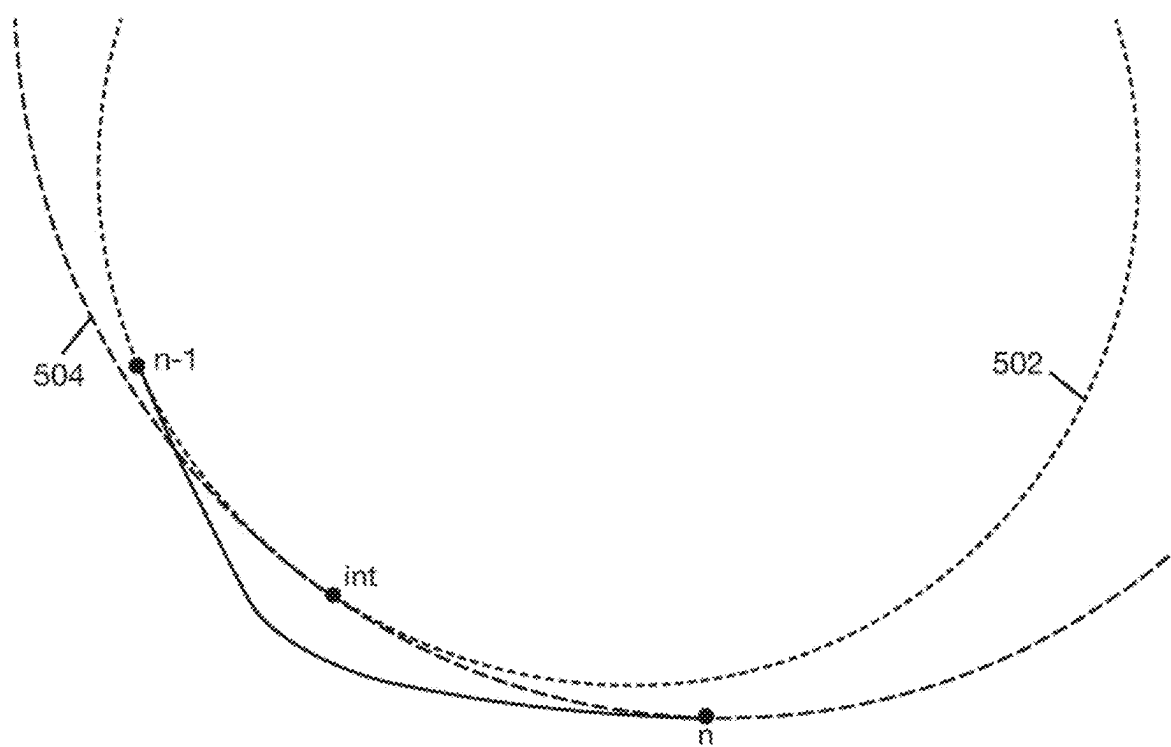


FIG. 5

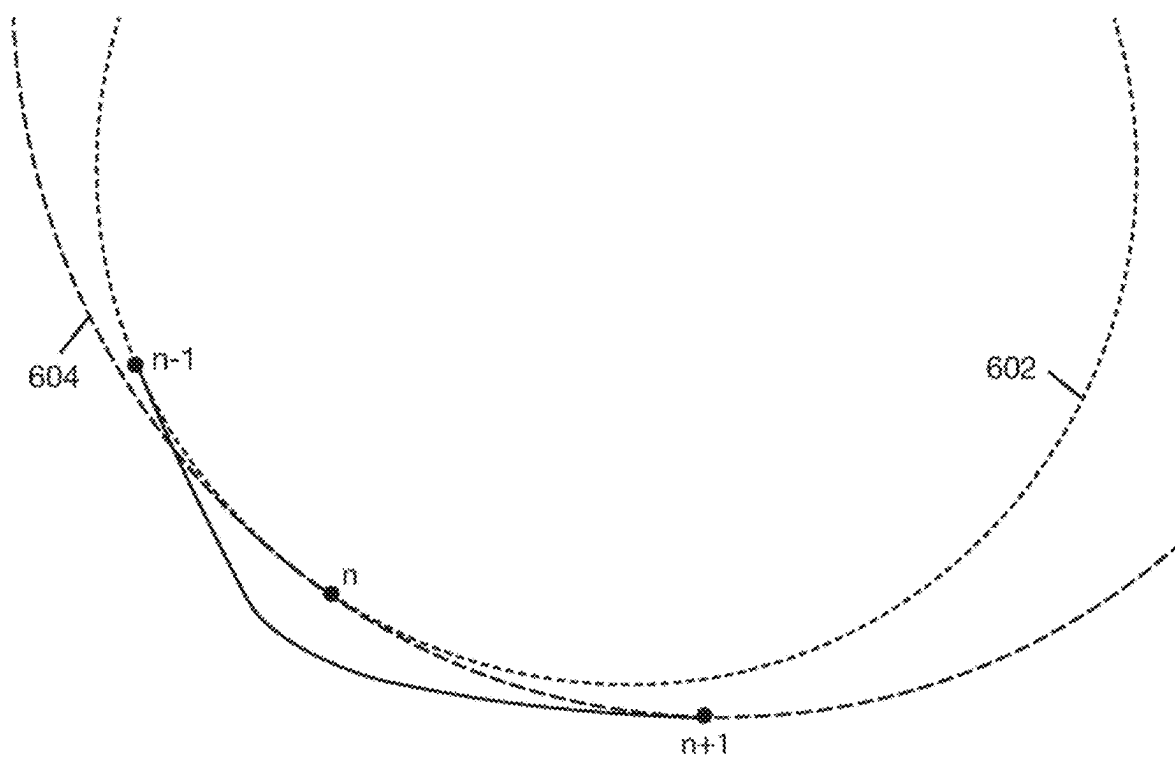


FIG. 6

1

METHOD OF DRILLING A WELLBORE TO A TARGET

CROSS-REFERENCES TO RELATED APPLICATIONS

This application is a continuation application of and claims the benefit of priority of U.S. patent application Ser. No. 16/097,788 filed on Oct. 30, 2018, entitled "METHOD OF DRILLING A WELLBORE TO A TARGET", which is a U.S. national phase application and claims the benefit of priority of application No. PCT/US2017/032295, filed on May 11, 2017, entitled "METHOD OF DRILLING A WELLBORE TO A TARGET", which claims the benefit of priority to U.S. Provisional Application No. 62/408,622, filed on Oct. 14, 2016, entitled "MEASUREMENT WHILE DRILLING METHOD", and U.S. Provisional Application No. 62/335,078, filed on May 12, 2016, entitled "METHOD FOR CORRECTING POSITIONAL OR VERTICAL DEPTH ERRORS IN DOWNHOLE DIRECTIONAL SURVEYS". Each of the foregoing applications in this paragraph are incorporated herein by reference as if fully set forth in their entireties for all purposes.

BACKGROUND OF THE INVENTION

In wellbore placement by measurement-while-drilling (MWD), surveys of the inclination and azimuth are taken at regular intervals along the wellbore. Industry standards require that a survey is taken at least every 100 feet. Commonly, a survey is taken every 90 feet, corresponding to the length of a "stand" consisting of three "singles" of drill pipe. In sections with strong curvature ("build-section"), it is common to take surveys every single of drill pipe (30 feet). The wellbore trajectory is then computed by minimum curvature interpolation, which implicitly assumes a circular arc (constant radial arc) between any two adjacent survey stations. Since the true well path between adjacent survey stations is not usually a circular arc, the actual change in position from one survey station to the next is different from the positional change computed by minimum curvature. This problem is well known in the industry. It is a cause of considerable concern because even small errors in vertical depth can lead to significant economic losses due to stranded hydrocarbon resources. To avoid such errors, methods are available to measure or estimate the direction of the wellbore in the intervals between MWD surveys.

One method used to identify wellbore trajectory between MWD surveys includes measuring the wellbore inclination and/or azimuth at many additional points between two stationary surveys. This is called "continuous survey" or "dynamic survey," even though in practice it just employs a higher sampling rate than the actual MWD survey stations. Continuous survey data may include inclination measurements only, or both inclination and azimuth measurements. The wellbore trajectory is then computed by minimum curvature interpolation that assumes a circular arc between any two adjacent points. The more additional data points there are between the stationary surveys, the closer the two adjacent points, and the more accurate the computed wellbore trajectory between the adjacent points may be. However, the use of additional intermediate data points to facilitate better calculation of wellbore trajectory introduces operational complications in many workflows that use directional survey data.

BRIEF SUMMARY OF THE INVENTION

In one aspect, a method of drilling a wellbore to a target includes measuring attitudes at two adjacent survey stations

2

along a wellbore using a downhole surveying tool. The method further includes determining an actual change in wellbore position over a survey leg linking the two adjacent stations. Corrections are then applied to the measured attitude at one or both of the survey stations such that a modeled well path joining the survey stations reflects the determined actual change in wellbore position between the two survey stations.

In another aspect, a method of drilling a wellbore to a target includes measuring attitudes at two adjacent survey stations along a wellbore using a downhole surveying tool. The method further includes determining an actual change in wellbore position over a survey leg linking the two adjacent stations. Then, a notional survey station is interpolated between the two adjacent survey stations. The interpolation involves computing the attitude of the notional survey station such that two modeled well paths linking the two adjacent survey stations and notional survey station reflect the determined actual change in wellbore position over the survey leg.

In yet another aspect, a method of drilling a wellbore to a target includes measuring attitudes at least at two of consecutive first, second, and third survey stations along a wellbore using a downhole surveying tool. An actual change in wellbore position between the first and third survey stations is determined. Then, corrections are applied to the attitude of the second survey station such that two modeled well paths linking the three survey stations reflect the determined actual change in wellbore position between the first and third survey stations.

BRIEF DESCRIPTION OF THE DRAWINGS

The following is a description of the figures in the accompanying drawings. The figures are not necessarily to scale, and certain figures and certain views of the figures may be shown exaggerated in scale or in schematic in the interest of clarity and conciseness.

FIG. 1 shows a system for drilling a wellbore.

FIG. 2 shows a well path with survey stations.

FIG. 3A is a flowchart illustrating a method of drilling a wellbore to a target, according to one embodiment.

FIG. 3B is a flowchart illustrating a method of drilling a wellbore to a target, according to another embodiment.

FIG. 3C is a flowchart illustrating a method of drilling a wellbore to a target, according to another embodiment.

FIG. 4A illustrates minimum curvature solution with standard survey.

FIG. 4B illustrates minimum curvature solution with corrected survey.

FIG. 5 illustrates a notional station interpolated between a previous and a current survey station.

FIG. 6 illustrates a notional station projected ahead of a current survey station.

DETAILED DESCRIPTION OF THE INVENTION

Measurement-while-drilling (MWD) survey stations are typically spaced about every 90 feet along the wellbore. Each MWD survey station is characterized by a measured depth, MD, inclination, Inc, and azimuth, Az. Inclination is the deviation of the wellbore from the vertical. Azimuth is the orientation of the wellbore relative to the north. Measured depth is the length of the wellbore. The survey data may be represented by a set of direction vectors D in a North-East-Vertical (NEV) coordinate system. Each direc-

3

tion vector D corresponds to a measured depth, MD, at a survey station and may be expressed as shown in Equation (1) below. (The direction vector D may also be described as the attitude of the corresponding survey station.)

$$D = |\sin(\text{Inc}) \cdot \cos(\text{Az}), \sin(\text{Inc}) \cdot \sin(\text{Az}), \cos(\text{Inc})| \quad (1)$$

where:

Inc=inclination

Az=azimuth

D=direction vector or attitude

The most widely used method for computing the well path in the industry is the minimum curvature method, which assumes a circular arc between any two adjacent survey stations with indices n-1 and n. The position coordinates of survey station n-1 are given or known. The position coordinates for survey station n, in terms of Easting, Northing, and True Vertical Depth, are calculated by fitting a circular arc to the two positional vectors. If the well path between two adjacent survey stations n-1 and n is represented by position vectors P_{n-1} and P_n , then

$$D_n = |\sin(\text{Inc}_n) \cdot \cos(\text{Az}_n), \sin(\text{Inc}_n) \cdot \sin(\text{Az}_n), \cos(\text{Inc}_n)| \quad (2A)$$

$$DL_n = \cos^{-1}(D_{n-1} \cdot D_n) \quad (2B)$$

$$P_n = P_{n-1} + (MD_n - MD_{n-1}) \cdot \tan\left(\frac{DL_n}{2}\right) / DL_n \cdot (D_{n-1} + D_n) \quad (2C)$$

where:

D_{n-1} =direction vector at survey station n-1

D_n =direction vector at survey station n

DL_n =dogleg angle between attitudes at survey stations n-1 and n

MD_{n-1} =measured depth at survey station n-1

MD_n =measured depth at survey station n

P_{n-1} =position vector at survey station n-1

P_n =position vector at survey station n

In the case where $DL_n=0$, the expression $\tan(DL_n/2)/DL_n$ in Equation (2C) is replaced by $1/2$.

The standard minimum curvature solution assumes that the entire interval between survey stations n-1 and n will be drilled at a constant radius of curvature. In reality, drilling of the interval will not follow a circular arc. In reality, the true changes in inclination and azimuth will not be distributed along a circular arc over the survey interval between the two adjacent survey stations. This means that the true change in position between the survey stations will be different from the change in position computed by the minimum curvature method (Equations (2A) through (2C)). It is possible to estimate and correct this positional error using additional information on the wellbore trajectory between surveys, such as through continuous surveying or slide sheets. However, in order for such a correction to be operationally viable, it has to fit within the standard wellbore surveying framework that employs minimum curvature representation of the wellbore trajectory.

In one embodiment of the present disclosure, a method of correcting wellbore positional error includes applying corrections to the measured attitude at one or both of two adjacent survey stations such that a modeled well path joining the two survey stations reflects the actual change in wellbore position (usually expressed in 3 coordinates, Northing, Easting, and True Vertical Depth) between the two adjacent survey stations. Let the two adjacent survey stations be a previous survey station, n-1, and a current survey station, n. As the terms "previous" and "current" are used, the previous survey station will have a smaller measured

4

depth (MD) compared to the current survey station. For this embodiment, the position vector, P_n , at the current survey station is estimated using additional survey information collected between the previous survey station, n-1, and the current survey station, n. Then, corrections to the attitude measured at the current survey station are computed such that the minimum curvature solution matches the estimated position at the current survey station, n. This is in contrast to the standard minimum curvature computation using uncorrected survey data, where the attitude at the current survey station is matched but additional knowledge regarding the position of the current survey station is ignored. FIG. 4A illustrates a standard minimum curvature solution with uncorrected survey 400. The actual well path is shown at 402. Note the error 408 in true vertical depth (TVD) between the trajectory computed by the standard minimum curvature solution 400 and the actual well path 402 at the current survey station, n. FIG. 4B illustrates a minimum curvature solution with corrected survey 412, according to the present disclosure. The actual well path is shown at 402. Note that the corrected survey has eliminated the TVD error at the current survey station, n, with the position of the wellbore trajectory computed by the minimum curvature solution matching the position of the actual well path at the current survey station, n.

For the embodiment described above, the modeled well path linking the two adjacent survey stations is described by Equations (3A) through (3C). Equation (3A) shows the direction vector, D_n , at the current survey station, n, with inclination correction, δInc_n , to the inclination measured at the current survey station, n, and azimuth correction, δAz_n , to the azimuth measured at the current survey station, n. In alternate examples, only the measured inclination or only the measured azimuth may be corrected. The term "correction to attitude" will generally mean correction to one or both of the measured inclination and azimuth. The dogleg angle, DL_n , and position vector, P_n , at the current survey station are given by Equations (3B) and (3C).

$$D_n = |\sin(\text{Inc}_n + \delta\text{Inc}_n) \cdot \cos(\text{Az}_n + \delta\text{Az}_n), \quad (3A)$$

$$\sin(\text{Inc}_n + \delta\text{Inc}_n) \cdot \sin(\text{Az}_n + \delta\text{Az}_n), \cos(\text{Inc}_n + \delta\text{Inc}_n)|$$

$$DL_n = \cos^{-1}(D_{n-1} \cdot D_n) \quad (3B)$$

$$P_n = P_{n-1} + (MD_n - MD_{n-1}) \cdot \tan\left(\frac{DL_n}{2}\right) / DL_n \cdot (D_{n-1} + D_n) \quad (3C)$$

where:

Az_n =azimuth at current survey station

Inc_n =inclination at current survey station

δInc_n =correction to inclination at current survey station

δAz_n =correction to azimuth at current survey station

D_{n-1} =direction vector, or attitude, at the previous survey station

D_n =direction vector, or attitude, at the current survey station

DL_n =dogleg angle between attitudes at the two survey stations

MD_{n-1} =measured depth at the previous survey station

MD_n =measured depth at the current survey station

P_{n-1} =position vector at previous survey station

P_n =position vector at the current survey station

In the correction in FIG. 4B, note that the well-path vector 410 is not tangent to curve 412 at station n. This is because the correction defines a new survey vector that is tangent to the desired curve, but is not necessarily aligned with the well

5

path. Relaxing this tangent restriction enables proper placement of the well using the circular arc. However, there is value in having the well path close to the tangent lines of the modeled curve 412. The embodiments described below use dual arc optimization to achieve this.

In another embodiment of the present disclosure, a method of correcting wellbore positional error includes applying corrections to the attitude at a notional station interpolated between two adjacent survey stations such that two modeled well paths joining the notional station to each of the two adjacent survey stations reflect the actual change in wellbore position between the two adjacent survey stations. For this embodiment, it is helpful first to consider the effect of interpolating a notional station between two adjacent survey stations, i.e., a previous survey station, $n-1$, and a current survey station, n , and modeling the interval between the two adjacent survey stations as two circular arcs. If the attitude of the interpolated notional station matches the attitude of the single circular arc linking the two adjacent survey stations, then the final position computed as two circular arcs remains unchanged. If the attitude of the interpolated notional station is changed slightly up or down, left or right, then the final position of the current survey station, n , computed from two circular arcs moves in the same direction. Thus, by selecting the correct attitude for the interpolated notional station, the position of the current survey station, n , as computed by two circular arcs can be made to closely match the estimated position at the current survey station, n . As in the previous embodiment, the position vector, P_n , at the current survey station, n , can be estimated using additional survey information collected between the previous survey station, $n-1$, and the current survey station, n .

For illustration purposes, FIG. 5 shows an example of a notional station, int, interpolated between two adjacent survey stations, $n-1$ and n . In FIG. 5, the curve 504 is tangent to (or defined by) the tangent lines defined by survey vectors (D) at stations int and n , and curve 502 is tangent to (or defined by) the tangent lines defined by survey vectors at stations int and $n-1$.

Suppose that the notional station, int, interpolated between two adjacent survey stations is at a measured depth MD_{int} with attitude D_{int} , then the two circular arcs (e.g., 502, 504 in FIG. 5) joining the adjacent survey stations to the notional station are solved by:

$$DL_1 = \cos^{-1}(D_{n-1} \cdot D_{int}) \quad (4A)$$

$$DL_2 = \cos^{-1}(D_{int} \cdot D_n) \quad (4B)$$

$$(4C)$$

$P_n =$

$$P_{n-1} + (MD_{int} - MD_{n-1}) \cdot \frac{\tan\left(\frac{DL_1}{2}\right)}{DL_1} \cdot (D_{n-1} + D_{int}) + (MD_n - MD_{int}) \cdot \frac{\tan\left(\frac{DL_2}{2}\right)}{DL_2} \cdot (D_{int} + D_n) + \Delta MD \cdot D_n$$

where:

DL_1 =dogleg angle between attitudes at the previous survey station and notional station

DL_2 =dogleg angle between attitudes at the notional station and current survey station

D_{n-1} =direction vector, or attitude, at the previous survey station

6

D_{int} =direction vector, or attitude, at the notional station
 D_n =direction vector, or attitude, at the current survey station

MD_{n-1} =measured depth at the previous survey station

MD_{int} =measured depth at the notional station

MD_n =measured depth at the current survey station

ΔMD =residual error in along-hole depth

P_{n-1} =position vector at previous survey station

P_n =position vector at the current survey station

Since the final equation is in vector form, it provides three equations, which can be solved for ΔMD and D_{int} . D_{int} is a unit vector with only two independent unknowns, which can alternately be represented as Inc_{int} and Az_{int} . The position of MD_{int} would normally be selected to be midway between MD_{n-1} and MD_n , in order to minimize the curvatures of the segments. The mathematics can be made simpler and a non-iterative solution can be found, without significantly degrading accuracy, by replacing each of the two expressions $\tan(DL/2)/DL$ with $1/2$.

The embodiment described above may be used if it is acceptable to add interpolated stations to the survey data or if it is acceptable to adjust the attitudes of alternate survey stations only. However, more commonly the preferred solution adjusts the attitude of each survey station in real time without adding additional stations.

In another embodiment of the present disclosure, a method of correcting wellbore positional error includes modeling the interval between three consecutive survey stations—a first station, $n-1$, a second station, n , and a third station, $n+1$ (where MD of the first station $< MD$ of the second station $< MD$ of the third station)—as two circular arcs, and applying corrections to the attitude of the second station, n , such that the two circular arcs reflect the actual change in wellbore position between the first station, $n-1$, and the third station, $n+1$. In one example, the stationary surveys at the three stations may be available, and the corrections to the second station may be determined using a procedure similar to the one described above for the interpolated notional station—in this case, the second station will correspond to the interpolated notional station, the first station will correspond to the previous station before the interpolated notional station, and the third station will correspond to the current station after the interpolated notional station. In another example, the calculation must be made knowing the well path prior to the second (or current) station, n , to be adjusted, but without knowledge of the third (or next) station, $n+1$. The most likely solution is therefore found by projecting the well path ahead to a notional station described by D_{n+1} and P_{n+1} in Equations (5A) and (5B), respectively, and adjusting the directional vector D_n using the above technique over the three stations at $n-1$, n , and $n+1$. Note that Equation (5A) sets the directional vector at the notional station (or third station), $n+1$, to be the same as the directional vector at the second station, n . In other examples, a different relationship between the directional vector at the notional (third) station and the second station may be used.

$$D_{n+1} = D_n \quad (5A)$$

60

$$P_{n+1} = P_n + (MD_n - MD_{n-1}) \cdot D_n \quad (5B)$$

where:

D_{n+1} =direction vector, or attitude, at the notional, or third, station

D_n =direction vector, or attitude, at the second, or current, station

P_n =position vector at the second, or current, station

P_{n+1} =position vector at the notional, or third, station
 MD_n =measured depth at the second, or current, station
 MD_{n-1} =measured depth at the first, or previous, station

In the example described above without knowledge of the third (or next) station, corrections may be applied to the current survey such that corrections needed to future surveys are minimized. In this case, the distance of the wellbore ahead of the second station, n , has not been surveyed yet, but it is possible to estimate the position of the next station (or third station) using secondary data, i.e., supplemental information that is outside of the stationary survey data (such as information from a qualitative orientation tool or sensor placed near the bit or information about how the well was directionally controlled during the drilling, e.g., depths and toolface directions information from slide sheets). This additional information can be used to create a more stable correction at the current station. Thus station “ $n-1$ ” is the previous (or first) survey station (used for the beginning of the traditional minimum curvature calculation), station “ n ” is the current (or second) survey station (used as the end point for traditional minimum curvature), and station “ $n+1$ ” is next (or third) survey station in the future whose position would have to be estimated. By incorporating this information into the solution used at the second (or current) survey station, n , the amount of correction that will be needed when the drill bit finally drills far enough to allow a stationary survey to be taken at the next survey station ($n+1$) will be reduced. Once the position of the notional future survey station ($n+1$) has been estimated, then the minimum curvature solution will be similar to the previous embodiment with three stations.

For illustration purposes, FIG. 6 shows a third station, $n+1$, projected ahead of a second (or current) station, n . The attitude of the second station, n , has been corrected such that the circular arcs (602, 604) linking the second station to the first (or previous) station, $n-1$, and third station, $n+1$, reflect the estimated actual change in wellbore position between the first and third stations. In FIG. 6, curve 604 is tangent to (or defined by) the tangent lines defined by survey vectors (D) at stations n and $n+1$, and curve 602 is tangent to (or defined by) the tangent lines defined by survey vectors (D) at stations at $n-1$ and n .

FIG. 1 shows an example of a drilling environment in which the embodiments described above may be used. In the example drilling environment, a drill string 100 including a bottom hole assembly (BHA) 102 is inserted through a wellhead 104 into a wellbore 106. The drill string 100 may be supported by a derrick assembly 110, as is well known in the art. The BHA 102 includes a drill bit 108 for drilling the wellbore 106. In one embodiment, the BHA 102 further includes a measurements section 112, which includes sensors and other equipment for making survey measurements from the wellbore 106. In one example, the measurements section 112 may include one or more downhole surveying tools, such as MWD module 114 and logging-while-drilling (LWD) module 116. The measurements section 112 may further include an electronics module 118, which may include a processor and other related computing devices for processing and applying corrections to survey data downhole and storing data. The measurements section 112 may include a communications module 120 for transmitting corrected and/or uncorrected survey data to a recording unit 122 at the surface. The recording unit 122 may be connected to appropriate computing facilities at the surface that allow processing of survey data received from the measurements section 112. The electronics module 118 and communications module 120 may be integrated into any or both of the

measurement modules 114, 116. The BHA 102 may include other tools for directional drilling, such as a rotary steerable system (RSS) 124. The RSS 124 may also include devices for making downhole measurements.

In accordance with the present disclosure, a method of drilling a wellbore, such as wellbore 106, to a target involves making survey measurements along the wellbore. The survey measurements may be made using any known MWD and/or LWD techniques known in the art. The method may start at any point in the wellbore with a known position and a known (or assumed) orientation. For example, the starting point could be at the wellhead, or at a kickoff point, i.e., where deflection of the wellbore from the vertical starts, or at the bottom of a casing in the wellbore, or at any other desired point in the wellbore with a known position. For illustration purposes, FIG. 2 shows an example well path 200 from a wellhead A to a target B. The starting point, S_0 , may be at the wellhead A or at another point along the well path. Subsequent survey stations, $S_1, S_2, \dots, S_{k-1}, S_k$, are located along the well path. The number of survey stations between the starting point, S_0 , and the target B will depend on the length of the wellbore between these points. The spacing between these survey stations will typically be around 90 feet (equivalent to the length of three drill pipes), but may be as small as 30 feet (equivalent to the length of a single drill pipe) in sections of the well path with strong curvature. The spacing between survey stations may be coordinated with the length of the drill pipe because stationary surveys can be taken when drilling is paused to allow addition of drill pipes to the drill string in the wellbore. Such stationary surveys are typically less noisy than surveys taken while drilling. The method may include drilling from one survey station to the next and making a measurement at each survey station after such drilling. The method may also include making additional survey measurements while drilling from one survey station to the next. The method includes modifying the measured attitudes at survey stations to ensure that modeled well paths between adjacent survey stations reflect true change in wellbore position between the adjacent survey stations, as described above. The method, as described herein, enables positional accuracy equivalent to the ones achievable by high-accuracy methods, such as described in the background, without the need to integrate numerous additional data into the survey record. The corrected attitudes may be used to define the trajectory of the wellbore and enable accurate steering of the wellbore to the target. Alternately, the corrected attitudes may be used for characterizing the well path after drilling.

FIG. 3A is a flowchart illustrating one practical implementation of a method of drilling to a target according to the present disclosure. The process may start at 300 at any point in the wellbore with a known position and a known (or assumed orientation). That known point where everything begins is the tie-in survey. The process may use an index n to keep track of the current survey station. In the next step 302, the index n is incremented by 1, and a downhole surveying tool, e.g., MWD module 114 in FIG. 1, advances to the next survey station along the wellbore. In one embodiment, advancing the downhole surveying tool to the next survey station may involve drilling the wellbore to the next survey station. In the next step 304, a new survey measurement is made at the current survey station n using the downhole surveying tool. In the next step 306, the actual change in position between the previous survey station, $n-1$, and the current survey station, n , is estimated using survey information available between the previous survey station and the current survey station. Some examples of how the

actual change can be estimated will be discussed below. In the next step 308, attitude corrections for the current survey leg are computed such that the minimum curvature trajectory from the previous survey station n-1 to the current survey station n reflects the actual change in wellbore position between the two stations. Variations to the workflow in FIG. 3A are possible depending on the wellbore positional error correction method used in the workflow. In general, any of the wellbore positional error correction methods described above may be used. FIGS. 3B and 3C show examples of variations to the workflow of FIG. 3A. In FIG. 3B, step 308B (corresponding to step 308 in FIG. 3A) is based on applying corrections to a notional station interpolated between two survey stations. In FIG. 3C, step 308C (corresponding to step 308 in FIG. 3A) is based on applying corrections to the current survey station taking into account a future survey station, and the preceding step 306C (corresponding to step 306 in FIG. 3A) involves estimating an actual change in position between the previous station and a future notional station determined by projecting the wellbore path ahead of the current survey station. Another variant not shown in the drawings may involve repeating step 304 such that three actual surveys are available as input for the wellbore positional error correction method in step 308C. That is, instead of using a future notional station, an actual third station where a survey measurement is taken using a downhole surveying tool may be used.

Returning to FIG. 3A, in the next step 310, the position of the current survey station, n, is compared to the target to see if the drill bit is at the target. If the drill bit is not at the target, steps 302 to 310 are repeated. In one embodiment, before repeating steps 302 to 310, steering parameters for a wellbore section to be drilled is determined using the corrected attitudes from step 308. As mentioned above, step 302 may involve drilling the wellbore to the next survey station. The corrected attitudes of step 308 may be used to determine a wellbore trajectory that accounts for actual change in wellbore position between survey stations, which can be used to make steering decisions while drilling the wellbore to the next survey station according to step 302. On the other hand, if the drill bit is at the target, the corrected projection to bit, i.e., corrected distance to drill the bit from the current survey station, is computed at 312. Corrected projection to bit, P_{bit} , from the current survey station, n, can be computed by extrapolation as follows:

$$P_{bit} = P_n + DTB \cdot D_n \quad (6)$$

where:

P_{bit} =corrected projection to bit

P_n =position vector at the current survey station n

DTB=distance from the surveying tool to the front of the drill bit

D_n =direction vector, or attitude, at the current survey station

The correction to bit is used to estimate the position of the hole that has been drilled but where there might not be survey measurements. Due to the presence of other drilling tools in the wellbore, the surveying instrument does not make it all the way down to the bottom of the wellbore. Therefore, it is necessary to estimate what the orientation of this final segment of the wellbore would be.

Steps 306 through 310 (or variants thereof) may be carried out in an appropriate module in the BHA, or parts or all of steps 306 through 310 (or variants thereof) may be

carried out with appropriate processing unit(s) or computer(s) at a surface location.

Determining Actual Change in Position Using Continuous Survey Data Method

According to one embodiment, the actual change in position from the previous survey station to the current survey station, n, in step 306 may be estimated from continuous survey measurements. The continuous survey measurements are taken while drilling from the previous survey station, n-1, to the current survey station, n. Although the survey is described as continuous, in practice it just means that the continuous survey employs a higher sampling rate than the actual survey stations (or provides additional survey data points between the survey stations). The continuous survey is separate from the stationary (or static) survey measurements that are made at survey stations (step 304). The continuous survey measurements typically include measured depth, inclination, and azimuth data. The capability to make the continuous survey measurements may be provided by any suitable tool, such as a continuous inclination survey tool, in the BHA.

For each survey interval from the previous survey station, n-1, to the current survey station, n, the following information is needed: continuous inclination and/or azimuth sequence $D(md_m)$, where $MD_{n-1} \leq md_m \leq MD_n$, $m=1 \dots M$, where MD is measurement depth. To compute the actual change in wellbore position, outliers from the M values of $D(md_m)$ are removed. Then, the corresponding change in position from the previous station, n-1, to the current station, n, is computed. Data outliers can be removed by low-pass filtering, resampling, spline fitting, or by other means known in the art. The output of this computation is the change in position from the previous station, n-1, to the current station, n, denoted $P_n - P_{n-1}$.

Pre-Processing of Continuous Survey Data

It should be noted that the continuous survey data may be subject to additional error sources not present in stationary MWD survey. Vibration during the drilling process may introduce noise to the measurement. In extreme cases, drilling tools such as agitators may be used that intentionally induce axial vibrations by converting energy from mud flow into linear motion. Single sensor readings will have increased susceptibility to residual calibration errors on the sensor being used for inclination measurements, as well as potential biases from assumed values of total gravity references. These error sources can cause discrepancies between the static MWD surveys and the data obtained via continuous inclination. To enable accurate estimation of the actual change in position between two adjacent stations, it is necessary to process the continuous inclination data prior to its use in the correction method described above so that the benefits of including additional curvature in the survey are not outweighed by the detriments of including poor quality data.

In one embodiment, pre-processing of continuous survey data (measured depth, inclination, azimuth) includes data conditioning, applying adjustments to the depth data, and applying adjustments to the continuous orientation data.

Data Conditioning

In one embodiment, statistical outliers are removed from the continuous survey dataset. Noise-reduction functions are

11

also applied to the dataset. This may include applying a smoothing function, resampling the data at a more convenient rate for analysis, or creating synthetic data as needed. The output of this step is a new continuous survey dataset (of the form measured depth, inclination, and azimuth) that is more amenable to analysis.

Adjusting Depth Data

The measured depth data is adjusted to better correlate with the stationary survey depths. This process may include evaluating drilling parameters that can correlate sensor depth to survey depths (such as bit depth, pump pressure, block position, and slide-rotate patterns), and it may include looking at residuals when subtracting the stationary survey data from the continuous survey data. These adjustments may be made across the entire dataset or across smaller subsets of the dataset as small as a single point.

Adjusting Continuous Orientation Data

The continuous orientation data (inclination and azimuth) is adjusted to better correlate with the stationary survey depths. Where discrepancies exist, the stationary survey will be assumed to be of superior quality to the continuous data. For inclination, this may include calculating an offset across the whole set or data or calculating multiple offsets to be applied to subsets of data as small as individual points. For azimuth data, the same treatment that is applied to the inclination data may be applied or the azimuth may be replaced by taking interpolated azimuths from the stationary survey set.

Determining Actual Change in Position between Stations Using Slide Sheet Method

A significant source of true vertical depth (TVD) errors is due to the widespread use of mud motors in directional drilling. Mud motors make use of a bend in the BHA. Consider a bend that causes a wellbore curvature of 5 degrees per 100 feet. If the drill string is rotated, the bend rotates in all directions, resulting in a straight hole or a small curvature in a direction which results from gravity and the BHA design. If on the other hand the drill string does not rotate, but the motor is driven by the mud flow, this results in a curve of 5 degrees per 100 feet. By setting the orientation ("toolface") of the bend, the curvature of the wellbore can be oriented in the desired direction. In practice, this means that the wellbore is a sequence of "rotate" sections in which the direction is constant or slightly curved in a fixed direction and curved "slide" sections in which the direction changes in the direction of the toolface setting. These alternating slide and rotate sections are not accounted for in the standard minimum curvature representation of the wellbore trajectory.

According to the present disclosure, in another embodiment, for step 306, the actual change in position from the previous survey station, $n-1$, to the current survey station, n , may be computed from slide sheets. A slide sheet is a record of whether the well was intentionally deviated ("sliding" with a motor) or whether it was drilled with the assumption that the path would be straight (rotary drilling). By using a record of the intended steering directions, the curvature of the wellbore between the stationary survey stations can be estimated.

For each survey interval from the previous survey station, $n-1$, to the current survey station, the following information

12

from the slide sheets is needed: (1) "Slide" or "Rotate" mode for each segment of the interval drilled in a single mode; (2) the measured depths at each of the M switch-overs between slide and rotate segments for the interval, and (3) the toolface reported for each "Slide" segment.

To compute the actual change in position, each sliding or rotating segment is modeled by a circular arc. Each circular arc is defined by its starting and ending depths MD_{m-1} and MD_m , starting attitude D_{m-1} , starting toolface direction T_{m-1} , and rate of curvature C_m . The ending attitude of a segment D_m can be computed from Equations (7A) to (7C). The final direction vector D_m is found by rotation of the initial wellbore vector D_{m-1} and unit vector Y_{m-1} about the pole by angle R_m .

$$R_m = C_m \cdot (md_m - md_{m-1}) \quad (7A)$$

$$Y_{m-1} = |\cos(I_{m-1}) \cdot \cos(A_{m-1}) \cdot \cos(T_{m-1}) - \sin(A_{m-1}) \cdot \sin(T_{m-1}) \cdot \cos(I_{m-1})| \cdot \cos(A_{m-1}) \cdot \cos(T_{m-1}) + \sin(A_{m-1}) \cdot \sin(T_{m-1}) \cdot \sin(I_{m-1}) \cdot \cos(T_{m-1}) \quad (7)$$

$$D_m = D_{m-1} \cdot \cos(R_m) + Y_{m-1} \cdot \sin(R_m) \quad (7C)$$

where

R_m = rotation angle

C_m = rate of curvature

md_m = ending measured depth

md_{m-1} = starting measured depth

Y_{m-1} = unit vector normal to the initial wellbore vector D_{m-1} and normal to the pole about which the wellbore interval curves

I_{m-1} = Starting inclination

A_{m-1} = Starting azimuth

T_{m-1} = starting toolface direction

D_m = ending attitude of a segment

D_{m-1} = starting attitude of a segment

Given values for each of the parameters, final direction vectors can be found for all of the segments within an interval by solving sequentially from the first segment to the last. Final position vectors P_m can be obtained by solving the circular arc formulas for each segment, the last such vector giving the current position P_n at the end of the interval.

Several different computation modes may be employed, depending on the nature of the survey interval.

Sliding Segments Only Mode

If the survey interval consists of a number of sliding segments but no rotating segments, all of the depths and toolfaces are obtained from slide sheet data. The only unknown is the rate of curvature C_M , which can be assumed to be the same during each sliding segment. For any curvature value C_M , a solution can be found for the final attitude vector of the interval D_M . The curvature value C_M which results in a final attitude vector closest to the measured attitude D_n at the end of the interval is taken to be the solution. The optimum value for C_M may be found by methods known in the art, such as Newton-Raphson iteration.

Sliding and Rotating Segments Mode

If the survey interval contains both sliding and rotating segments, the curvatures of each mode are unknown. It may be assumed that all sliding segments have the same curvature C_S , and all rotating segments have the same curvature C_R . If the rotating segments are assumed to have toolface zero (rotating mode curvature is restricted to the vertical

13

plane), the optimum values of C_S and C_R will match the final attitude exactly. In this case the sliding mode matches all left/right deviation in the horizontal plane, and the rotating mode curvature matches the residual up/down deviation in the vertical plane. As the computation is non-linear, iterative methods are again required to find the solution.

If the rotating mode segments are allowed to deviate out of vertical, then the rotating mode toolface T_R represents a third unknown. In this case the problem cannot be solved using data from a single interval or stand, as the final measured attitude provides only two independent equations. It can be solved using data from the current interval and the previous interval, finding the parameters which best fit to both measured attitudes D_{n-1} and D_n .

For actively controlled steering tools which generate different curvatures in different segments within one interval, the ratio of segment curvatures should be provided (e.g., one segment at 50% and another at 100%).

There are certain limitations to these techniques. If all sliding toolfaces and rotating toolface coincide, it is not possible to find a unique solution. Under these circumstances one additional piece of information must be provided, such as either the sliding or rotating curvature.

It should be noted that slide sheets report one toolface per segment. A segment drilled with constant toolface does not generally follow a circular arc; however, the solutions provided above model each segment as a circular arc defined by its initial toolface direction. In order to limit systematic errors, the toolface reported on the slide sheet can be assigned to the midpoint of the corresponding circular arc. It is then possible to compute a starting toolface for use in the above calculations.

While the invention has been described with respect to a limited number of embodiments, those skilled in the art of, having benefit of this disclosure, will appreciate that other embodiments can be devised which do not depart from the scope of the invention as disclosed herein. Accordingly, the scope of the invention should be limited only by the accompanying claims.

What is claimed is:

1. A method of drilling a wellbore to a target, comprising: receiving, from a computer system, attitudes at a first survey station and a second survey station along a wellbore path using a downhole surveying tool, wherein the first survey station and the second survey station are consecutive;

modeling, by the computer system, based on a first attitude of the wellbore at the first survey station, a first interval between the first survey station and the second survey station as a first modeled well path;

determining, by the computer system, based on the first modeled well path, a first positional vector of the wellbore at the first survey station;

modeling, by the computer system, based on a second attitude of the wellbore at the second survey station, a second interval between the second survey station and an estimated position of a third survey station as a second modeled well path;

determining, by the computer system, based on the second modeled well path, a second positional vector of the wellbore at the second survey station;

estimating, by the computer system, a position of the third survey station based on the first modeled well path, second modeled well path, first positional vector, and the second positional vector, wherein the third survey station is beyond the second survey station along the wellbore path; and

14

drilling in accordance with the estimated position of the third survey station.

2. The method of claim 1, wherein estimating the position of the third survey station comprises determining a third attitude of the wellbore at the third survey station.

3. The method of claim 2, wherein determining the third attitude of the wellbore at the third survey station comprises setting the third attitude to be the same as the second attitude of the wellbore at the second survey station.

4. The method of claim 1, the method further comprising: applying corrections to the second attitude of the wellbore at the second survey station based on drilling in accordance with the estimated position of the third survey station.

5. The method of claim 1, wherein estimating the position of the third survey station further comprises using supplemental information outside of stationary survey data.

6. The method of claim 1, the method further comprising: after estimating the position of the third survey station, applying a minimum curvature solution to the first interval and the second interval such that the first modeled well path and the second modeled well path intersect at the second survey station.

7. The method of claim 1, wherein a first measured depth of the first survey station is less than a second measured depth of the second survey station, and the second measured depth is less than a third measured depth of the third survey station.

8. The method of claim 1, wherein the third survey station is one of a notional survey station, an actual survey station, or a planned survey station according to a planned wellbore path.

9. The method of claim 1, wherein the second positional vector of the wellbore at the second survey station is determined using a continuous survey method.

10. The method of claim 1, wherein the second positional vector of the wellbore at the second survey station is determined using a slide sheet method.

11. A non-transitory computer-readable medium comprising processor-executable instructions configured to cause one or more processors to:

receive attitudes at a first survey station and a second survey station along a wellbore path using a downhole surveying tool, wherein the first survey station and the second survey station are consecutive;

model, based on a first attitude of the wellbore at the first survey station, a first interval between the first survey station and the second survey station as a first modeled well path;

determine, based on the first modeled well path, a first positional vector of the wellbore at the first survey station;

model, based on a second attitude of the wellbore at the second survey station, a second interval between the second survey station and an estimated position of a third survey station as a second modeled well path;

determine, based on the second modeled well path, a second positional vector of the wellbore at the second survey station;

estimate, a position of the third survey station based on the first modeled well path, second modeled well path, first positional vector, and the second positional vector, wherein the third survey station is beyond the second survey station along the wellbore path; and

control one or more components of a drilling rig to drill in accordance with the estimated position of the third survey station.

15

12. The non-transitory computer-readable medium of claim 11, wherein the second positional vector of the wellbore at

the second survey station is determined using a continuous survey method.

13. The non-transitory computer-readable medium of claim 11, wherein the one or more processors are configured to execute further processor-executable instructions stored in the non-transitory computer-readable medium to:

determine a third attitude of the wellbore at the third survey station.

14. The non-transitory computer-readable medium of claim 13, wherein the one or more processors are configured to execute further processor-executable instructions stored in the non-transitory computer-readable medium to:

set the third attitude to be the same as the second attitude of the wellbore at the second survey station.

15. The non-transitory computer-readable medium of claim 11, wherein

the third survey station is one of a notional survey station, an actual survey station, or a planned survey station according to a planned wellbore path.

16. The non-transitory computer-readable medium of claim 11, wherein the one or more processors are configured to execute further processor-executable instructions stored in the non-transitory computer-readable medium to:

apply corrections to the second attitude of the wellbore at the second survey station based on drilling in accordance with the estimated position of the third survey station.

17. The non-transitory computer-readable medium of claim 11, wherein instructions to estimate the position of the third survey station further cause the one or more processors to execute further processor-executable instructions stored in the non-transitory computer-readable medium to:

receive supplemental information outside of stationary survey data; and

estimate the position of the third survey station based on the supplemental information.

18. The non-transitory computer-readable medium of claim 11, wherein the one or more processors are configured to execute further processor-executable instructions stored in the non-transitory computer-readable medium to:

16

after estimating the position of the third survey station, apply a minimum curvature solution to the first interval and the second interval such that the first modeled well path and the second modeled well path intersect at the second survey station.

19. The non-transitory computer-readable medium of claim 11, wherein a first measured depth of the first survey station is less than a second measured depth of the second survey station, and the second measured depth is less than a third measured depth of the third survey station.

20. A method of drilling a wellbore to a target, comprising:

measuring attitudes at two adjacent survey stations along a wellbore using a downhole surveying tool;

determining an actual change in wellbore position over a survey leg by linking the two adjacent survey stations with a first modeled well path; and

interpolating a notional survey station between the two adjacent survey stations, the interpolating comprising computing an attitude of the notional survey station such that a second modeled well path intersects the first modeled well path and the notional survey station, the notional survey station reflecting the determined actual change in wellbore position over the survey leg; and drilling in accordance with the notional survey station interpolated between the two adjacent survey stations.

21. The method of claim 20, wherein determining the actual change in wellbore position comprises determining the actual change using continuous survey measurements taken at least between the two adjacent survey stations during drilling of the wellbore.

22. The method of claim 20, wherein determining the actual change in wellbore position comprises determining the actual change using continuous inclination measurements taken at least between the two adjacent survey stations during drilling of the wellbore.

23. The method of claim 20, wherein determining the actual change in wellbore position comprises determining the actual change using depths and toolface directions defining slide intervals during drilling of the wellbore.

24. The method of claim 20, wherein the first modeled well path and the second modeled well path are circular arcs computed by a minimum curvature method.

* * * * *